

Mérida Urban Intelligence: A Geospatial Data Warehouse for Demographic, Economic and Public-Safety Analysis

Yazira Abyade Hernández Rives

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Repository: github.com/yazirahernandez/merida-urban-intelligence

Abstract. This project integrates four public sources (INEGI Census 2020, DENUE 05/2026, the INEGI Geostatistical Framework and SESNSP crime records) into a dimensional Data Warehouse implemented in PostgreSQL/PostGIS. Because the sources do not share a geographic representation, the core design problem was selecting a unit of analysis and resolving the spatial integration. The solution uses a two-scale model: 501 urban AGEs for demographic and economic analysis inside Mérida, and 106 municipalities for public safety, since no georeferenced incident data exist. All 14 required KPIs are computed from warehouse views. Spatial analysis shows strong clustering of age structure (Moran's $I = 0.84$) and schooling (0.78), a concentric commercial core that co-locates with dense residential rings (bivariate $I = 0.28$), and a regional crime cluster along the Mérida–northern coast corridor ($I = 0.26$).

1. Problem and Data Sources

The objective is a reproducible warehouse that allows territorial KPIs to be calculated, areas within Mérida to be compared, and geographic associations between demographic, economic and public-safety patterns to be tested. Table 1 summarises the four sources. Each arrives at a different grain and with a different spatial representation: tabular records with geographic keys (census), point coordinates (DENUE), polygons (framework) and municipal aggregates without coordinates (crime).

Table 1. Data sources

Layer	Source and edition	Original grain / representation	Key volume
Demographic	INEGI Census 2020, results by AGEb and urban block	Block, with AGEb, locality and municipal totals; CVEGEO keys, no geometry	1,532 AGEs, 106 municipalities in Yucatán; 32 variables used
Economic	INEGI DENUE, Yucatán, 05/2026	One establishment; latitude/longitude, SCIAN class, staff size	146,383 establishments; 56,909 in Mérida municipality
Geographic	INEGI Geostatistical Framework, Dec 2025	Urban AGEb and municipal polygons (ITRF2008 LCC)	1,562 AGEb and 106 municipal polygons
Public safety	SESNSP municipal crime incidence, 2015–2025	Municipality × year × subtype × modality, 12 month columns; no coordinates	114,268 rows; 157,917 incidents

2. Geographic Integration Strategy

Four candidate units were evaluated. **Urban blocks** were rejected because the census suppresses most indicators for blocks with one or two dwellings, and block-level business counts are too sparse. **Colonias** are not an official INEGI unit and cannot be joined to the census. The **municipality** collapses the city into a single observation. The **urban AGEb** was selected: it is the finest unit with unsuppressed census data, it has official polygons, and the locality of Mérida (31-050-0001) provides 501 units, enough for local spatial statistics.

No public dataset of georeferenced crime incidents exists for Mérida. Crime was therefore integrated at the municipal scale, producing a **two-scale design**: AGEb for demographic and economic analysis, and municipality for public safety. Municipal crime was deliberately *not* disaggregated to AGEs. Allocating it by population would give every AGEb the same crime rate, and allocating it by business counts would manufacture the very crime–business association the analysis intends to test. The latitude/longitude-to-polygon requirement was demonstrated with DENUE points instead (Table 2).

Table 2. Spatial integration results

Integration	Method	Result
Census → AGEb polygon	13-digit CVEGEO key	483 of 483 census AGEs matched (100%); 18 post-2020 polygons have no census data
DENUE → AGEb / municipality	ST_MakePoint(lon, lat) in EPSG:4326, then ST_Intersects in PostGIS	99.8% of Mérida-municipality establishments fall in an AGEb; 100% agree with the AGEb key reported by DENUE
Crime → municipality	5-digit CVEGEO key	106 of 106 municipalities matched

The framework polygons were validated (0 invalid geometries), areas were computed in EPSG:6372 (metric Lambert Conformal Conic) and geometries were stored in EPSG:4326. The independent agreement between the point-in-polygon result and DENUe's own AGEb attribute is the main evidence that the spatial integration is correct.

3. Data Warehouse Architecture

The pipeline follows RAW → CLEAN → SPATIAL JOIN → DW. A Python ETL reads the original ZIP files without modifying them, cleans and standardises the data, and loads it into a staging schema (stg). SQL scripts then build the dimensional model in the dw schema, perform the point-in-polygon join inside PostGIS and create the KPI views. The database runs in a Docker container (PostgreSQL 16, PostGIS 3.4), and the complete process is idempotent and rebuildable from the raw files. Figure 1 shows the model.

Merida Urban Intelligence - Modelo dimensional (PostgreSQL/PostGIS, esquema dw)

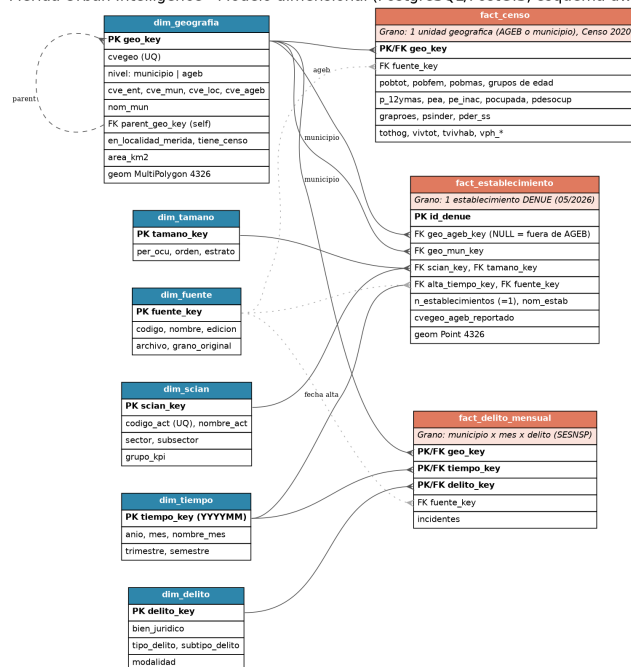


Figure 1. Dimensional model. Blue: dimensions; orange: fact tables with their grain. *dim_geografia* is a conformed dimension shared by all facts, with an AGEb → municipality hierarchy.

4. KPIs and Spatial Analysis Methods

All 14 required KPIs are computed in three SQL views: *v_kpi_ageb*, *v_kpi_municipio* and *v_delitos_tipo_tiempo*. Three definitional choices matter for interpretation. Retail uses SCIAN sector 46 and services SCIAN 51–81. The dominant activity is measured at the three-digit subsector level, because at two digits sector 46 dominates 402 of 501 AGEbS and the indicator loses discriminating power. Per-capita rates are withheld for AGEbS with fewer than 100 residents (16 units), where a handful of establishments produces extreme values.

Crime KPIs use the 2023–2025 annual mean, the period closest to the DENUe edition. Because small municipalities produce unstable rates (a municipality of 2,000 people can double its rate with two additional incidents), crime rates were smoothed with **Empirical Bayes**, which shrinks each rate towards the state mean in proportion to its population uncertainty. Global autocorrelation of the crime rate was also tested with the Assunção–Reis EB-standardised Moran's I.

Spatial-neighbourhood rule. Neighbours are defined by Queen contiguity: two units are neighbours if they share an edge or a vertex. Weights are row-standardised, so each spatial lag is the mean of the neighbours' values. Units without contiguous neighbours (one AGEb, two municipalities) were connected to their nearest neighbour. This rule has consequences: large peripheral AGEbS receive many neighbours (up to 20), while small central AGEbS receive few, so central units are compared with a narrower surrounding area. Density variables were log-transformed to reduce skew. Inference uses 999 conditional permutations with $\alpha = 0.05$. Spearman correlation was preferred to Pearson because of heavily skewed distributions.

5. Main Findings

5.1 Urban structure

The demographic maps (Figure 2) reveal two distinct spatial logics. Age structure is **concentric**: the share of residents aged 65 or older is highest in the historic centre and inner ring and declines outwards. Schooling follows a **north–south gradient**: average years of schooling are highest in the north and north-east and lowest in the southern periphery. Population density peaks in a ring of neighbourhoods around the centre, particularly to the south, west and east, rather than in the centre itself.

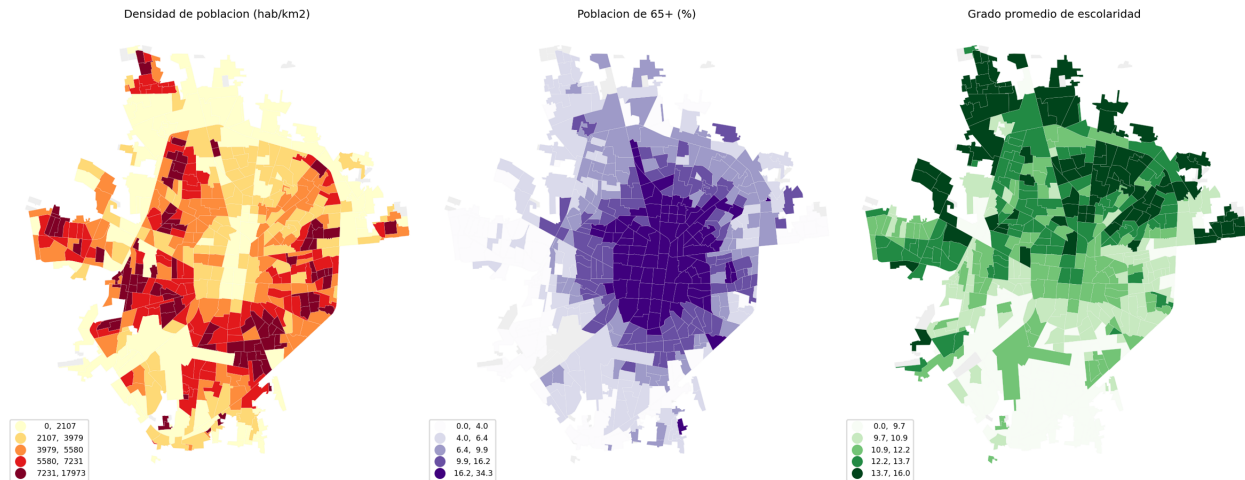


Figure 2. Population density, share of population aged 65+ and average years of schooling by AGEB (Census 2020, quintile classes; panel labels in Spanish as generated by the pipeline).

Economic activity (Figure 3) is concentrated in the historic centre and radiates outward along the consolidated city. The dominant subsector also changes with location: professional services (541) and ambulatory health services (621) dominate the central and north-central AGEBs, food and beverage preparation (722) is frequent in the north, and neighbourhood grocery retail (461) dominates most of the southern and western city.

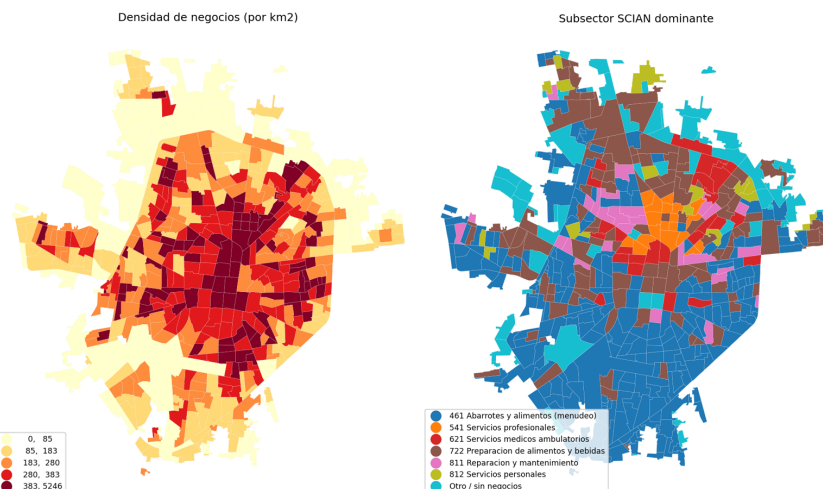


Figure 3. Business density and dominant SCIAN subsector by AGEB (DENUE 05/2026).

5.2 Global autocorrelation and correlation

Every AGEB indicator is significantly clustered (Table 3). The strongest patterns are demographic composition, not density: age structure ($I = 0.84$) and schooling ($I = 0.78$) show that Mérida is strongly segregated by life stage and educational level. Economic densities cluster at a moderate level (0.49–0.53). At the municipal scale, both the smoothed crime rate and business density are clustered.

Table 3. Global Moran's I (Queen contiguity, row-standardised)

Scale	Indicator	Moran's I	z	p (999 perm.)
AGEB (n = 465)	Share aged 65+	0.839	28.99	0.001
	Average years of schooling	0.777	27.19	0.001
	Economically active population rate	0.559	19.11	0.001
	Service density (log)	0.527	17.57	0.001
	Business density (log)	0.508	17.26	0.001
	Retail density (log)	0.488	17.18	0.001
	Population density (log)	0.468	15.98	0.001
Municipality (n = 106)	Business density (log)	0.285	4.95	0.001
	Crime rate, EB-smoothed	0.255	4.31	0.001
	Crime rate, EB-standardised Moran (Assunção-Reis)	0.269	4.57	0.001

Table 4 reports the correlation analysis together with the bivariate Moran's I, which relates a variable in each unit to a second variable averaged over its neighbours. Population density and business density are positively associated both within AGEBS ($\rho = 0.43$) and across neighbours (bivariate $I = 0.28$), so dense residential areas are adjacent to business-dense areas. The share of older residents correlates with retail density ($\rho = 0.38$), and the economically active population rate correlates negatively with businesses per resident ($\rho = -0.36$). These two results are best read together: the historic centre combines an older population, which lowers the active-population rate, with a high concentration of establishments. Age structure is a plausible confounder; the association does not imply that businesses reduce labour participation. Schooling and service density are unrelated ($p = 0.32$).

Table 4. Correlation (Spearman) and bivariate spatial association

Scale	Relationship	Spearman ρ	p	Bivariate I (x vs. neighbours' y)
AGEB	Population density vs. business density	0.434	<0.001	0.281 ($p = 0.001$)
AGEB	Share aged 65+ vs. retail density	0.383	<0.001	—
AGEB	Active population rate vs. businesses per 1,000	-0.356	<0.001	—
AGEB	Schooling vs. service density	-0.047	0.317	-0.028 ($p = 0.288$)
Municipality	Business density vs. crime rate (EB)	0.089	0.367	0.102 ($p = 0.057$)
Municipality	Population density vs. crime rate (EB)	0.056	0.569	0.172 ($p = 0.006$)
Municipality	Businesses per 1,000 vs. crime rate (EB)	0.065	0.509	—

5.3 Local spatial patterns

The LISA maps (Figure 4) locate these patterns. For business density, 85 AGEBS form a single high-high core around the historic centre with an extension to the east, while low-low clusters (36 AGEBS) lie along the northern and north-eastern urban edge and in the southern fringe. For population density, high-high clusters (76) form separate pockets in the southern, western and eastern inner ring, and low-low clusters (52) appear in the northern periphery and in a north-south strip north of the centre.

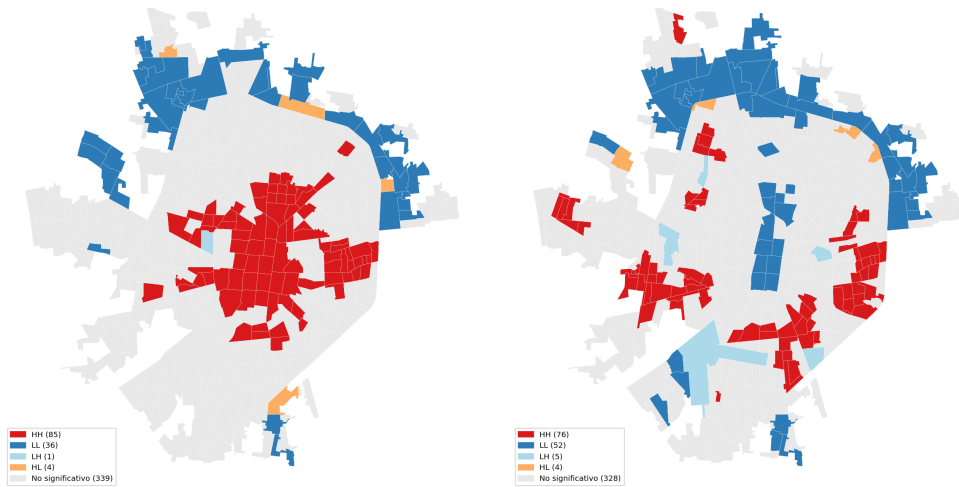


Figure 4. LISA clusters for business density (left) and population density (right), $p < 0.05$. HH: high surrounded by high; LL: low surrounded by low; LH/HL: spatial outliers.

LISA bivariado AGEB: densidad de población (x)
vs densidad de negocios de las AGEB vecinas (Wy)

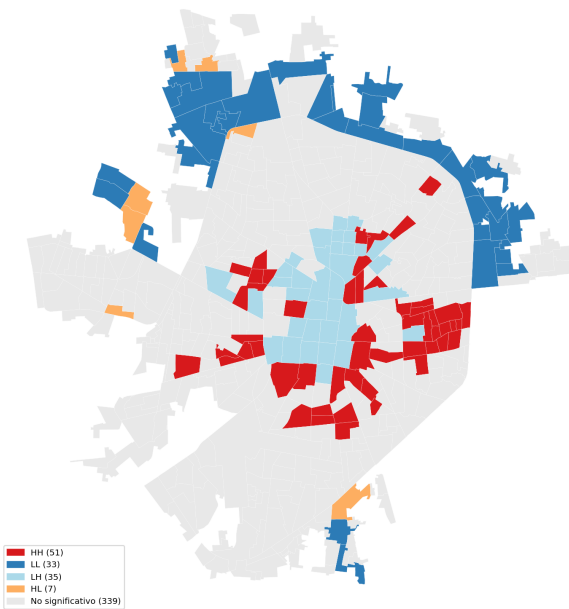


Figure 5. Bivariate LISA: population density of each AGEB (x) versus business density of its neighbours (Wy), $p < 0.05$.

The bivariate LISA (Figure 5) clarifies how people and businesses relate in space. The historic centre appears as low–high (35 AGEBs): relatively low residential density surrounded by high business density, consistent with a commercial centre that has relatively few residents. It is surrounded by a ring of high–high AGEBs (51), where dense residential areas border business-dense ones. The northern periphery is low–low on both variables.

5.4 Public safety

In Mérida municipality, 7,131 incidents were recorded in 2023–2025, an average of about 2,380 per year (2.4 per 1,000 residents). The most frequent offences were property damage (1,668), drug dealing (938), robbery (709), threats (655) and domestic violence (580); November was the month with most incidents and January the month with fewest. At state level, the smoothed crime rate forms a significant regional cluster (Figure 6, left). The high–high cluster is a corridor linking Mérida with the northern coast: Mérida, Progreso, Ixil, Motul, Baca, Mochochá, Bokobá, Dzemul, Telchac Pueblo and Suma. Low–low clusters occupy a large block of south-central and eastern municipalities.

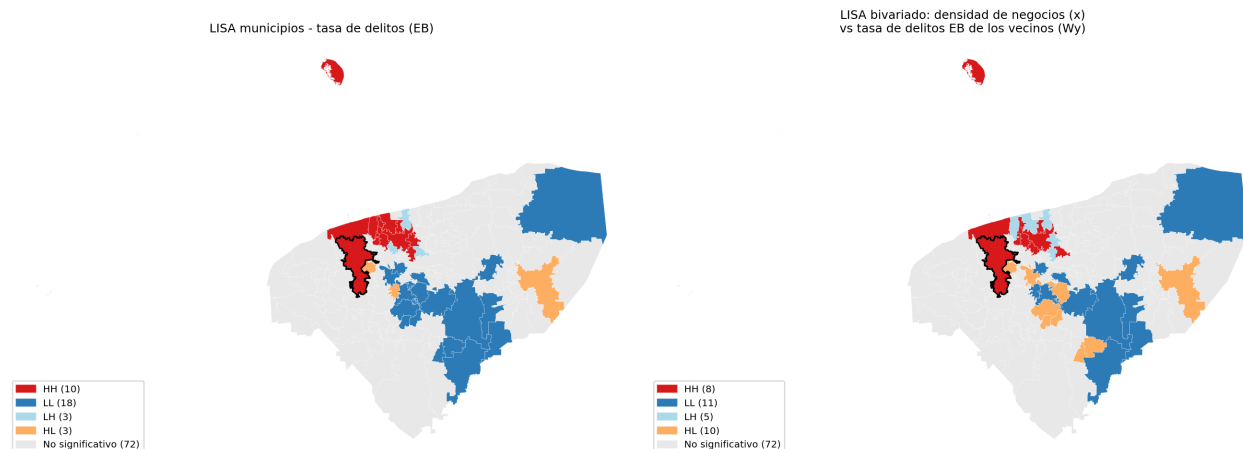


Figure 6. Left: LISA of the EB-smoothed crime rate by municipality. Right: bivariate LISA of business density (x) versus the neighbours' crime rate (Wy). Black outline: Mérida.

The municipal association between crime and urban intensity is regional rather than one-to-one. Within municipalities, the crime rate does not correlate with business or population density (Spearman ρ between 0.06 and 0.09, not significant). Pearson coefficients are higher (≈ 0.27 – 0.28), but they are driven by a few extreme values, which justifies the rank-based measure. Spatially, however, densely populated municipalities tend to border high-crime municipalities (bivariate $I = 0.17$, $p = 0.006$). The equivalent relationship for business density is marginal ($I = 0.10$, $p = 0.057$) and is not significant at $\alpha = 0.05$.

6. Limitations and Interpretation Cautions

- **No georeferenced crime.** Public safety can only be analysed across municipalities; no statement can be made about crime patterns inside Mérida, and crime cannot be related to AGEb-level indicators.
- **Recording quality.** SESNSP records are investigation files, not offences, and are subject to under-reporting. Recorded incidents in Yucatán fall from 34,716 (2015) to about 4,200 (2022–2024), an 88% drop that may reflect recording practices rather than a real decline; trends were therefore not analysed.
- **Temporal mismatch.** Census 2020, DENUE 2026 and crime 2023–2025 are combined; per-capita rates use 2020 populations.
- **Cartographic version.** The Dec 2025 framework includes 18 AGEbS created after 2020 that have no census data; they count for business densities but are excluded from per-capita KPIs.
- **Small numbers and coverage.** Rates for low-population units are unstable (addressed with a 100-resident threshold and EB smoothing). DENUE covers visible establishments and may under-represent informal or home-based activity.
- **MAUP and neighbourhood rule.** Results depend on the unit of analysis and on Queen contiguity; a different unit or a distance-based rule could change coefficient magnitudes and cluster boundaries.
- **Association is not causation.** All results describe co-location. Clustering can arise from shared historical, planning or socio-economic processes that were not modelled.

7. Conclusion

The project delivers a reproducible PostgreSQL/PostGIS warehouse that integrates four heterogeneous sources through a defensible two-scale geographic model and computes all required KPIs from warehouse views. The analysis shows that Mérida is structured concentrically by age and commercial activity and along a north–south axis by schooling, that residential and commercial intensity co-locate across neighbouring areas, and that recorded crime clusters regionally along the Mérida–northern coast corridor. The most valuable extension would be access to georeferenced incident data, which would allow public-safety indicators to enter the AGEb-level model.

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